(Please write your Enrollment Number) Enrollment No. 04801172022

MID-TERM EXAMINATION

B. Tech., Semester : VIIth

(October 2025) OFF LINE mode

Subject Code: BAI 403	Subject: Computer Vision
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) What is the goal of an optimization algorithm in machine learning?	2.5	1
	b) What happens if the learning rate in SGD is too large?	2.5	2
	c) What is parameter sharing, and where is it used?	2.5	1
	d) Why is the choice of loss function important for model performance?	2.5	2
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Explain gradient explosion and vanishing gradient problems with reasons and solutions.	5	2
	b) How weights are updated in Stochastic Gradient Descent algorithm?	5	1
	c) Explain the step-by-step process of training a neural network using forward pass, loss computation, and backward pass.	5	2
	UNIT II		CO Mapping
Q3	Attempt any two parts	(5*2=10)	
	a) Explain different types of layers used in CNNs: Convolutional, Pooling, Activation, and Fully Connected.	5	1
	b) How transfer learning be effectively utilized for domain-specific task?	5	2
	c) Explain ResNet architecture.	5	1